HybridNailClassifier: A Hybrid Deep Learning Framework for Non-Invasive Anemia Detection Using Nail Images

A REPORT

submitted by

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in partial fulfilment for the award

of

M. Tech. (Integrated) Software Engineering

School of Computer Science and Engineering

VIT
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School of Computer Science and Engineering

DECLARATION

I hereby declare that the project entitled **"HybridNailClassifier:A Hybrid Deep Learning Framework for Non-Invasive Anemia Detection Using Nail Images"** submitted by me to the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai Campus, Chennai 600127 in partial fulfilment of the requirements for the award of the degree of **M. Tech. (Integrated) Software Engineering** is a record of bonafide work carried out by me**.** I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or university.

Signature
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CERTIFICATE

The project report entitled "**HybridNailClassifier: A Hybrid Deep Learning Framework for Non-Invasive Anemia Detection Using Nail Images**" is prepared and submitted by **AFNAAN AHMED P (Register No: 22MIS1157)**. It has been found satisfactory in terms of scope, quality and presentation as partial fulfilment of the requirements for the award of the degree of **M. Tech. (Integrated) Software Engineering** in Vellore Institute of Technology, Chennai, India.

Examined by:

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Certificate of Participation

This is to certify that Mr/Ms. ………………………………..AFNAAN AHMED P (22MIS1157)……………………..……………………… of Vellore Institute of Technology, Chennai has successfully completed the **"Summer Research Internship on Smart Healthcare 2025"** organized by the Centre for Healthcare Advancement, Innovation and Research (CHAIR), Vellore Institute of Technology, Chennai from 15.05.2025 to 20.08.2025.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CNN	Convolutional Neural Network
YOLO	You Only Look Once
AUC	Area Under the Curve
ROI	Region of Interest
RGB-D	Red, Green, Blue – Depth
ReLU	Rectified Linear Unit
BN	Batch Normalization
CI	Confidence Interval
AdamW	Adaptive Moment Estimation with Weight Decay
VGG	Visual Geometry Group Network
ConvNeXt	Convolutional Network with Next-Generation Design
AHE	Adaptive Histogram Equalization
mAP	Mean Average Precision
FC	Fully Connected Layer

ABSTRACT

Anemia remains one of the most prevalent global health challenges, affecting over **1.6 billion individuals worldwide** [1], [2], particularly in low-resource regions where access to diagnostic facilities is limited. Conventional detection of anemia relies heavily on invasive blood tests that require laboratory infrastructure, trained medical personnel, and costly equipment [3]. These methods, while clinically accurate, are impractical for rural or economically disadvantaged populations where healthcare access is sparse. With the increasing penetration of smartphones and advancements in computer vision, **non-invasive and accessible solutions** have become a promising alternative for early anemia screening [4], [5].

However, most current non-invasive systems suffer from several limitations. They depend on controlled lighting conditions, specialized sensors, or uniform skin tones to maintain accuracy [6]. Traditional deep learning models also struggle with inconsistent illumination, varied image quality, and diverse skin pigmentation, leading to unreliable results [7], [8]. Moreover, earlier nail-based approaches such as **DenseNet- or VGG-based models** achieved only moderate accuracies (70–90%) and lacked real-time efficiency, making them unsuitable for smartphone deployment in field settings [6], [7], [9].

To address these challenges, this project introduces **HybridNailClassifier**, a hybrid deep learning framework for non-invasive anemia detection using nail images. The system integrates **YOLOv8** for precise nail region localization [10] and a **ConvNeXt-Tiny backbone** enhanced with a **Channel Attention module** for feature extraction and classification [11], [12]. A dataset consisting of **4,200 training and 852 patient-level test images**, encompassing diverse lighting, skin tones, and capture devices, was used for evaluation [6], [9], [13]. The model achieved an impressive **96% accuracy, 96% precision, 96% recall, and an AUC of 0.98**, outperforming existing methods while maintaining lightweight computational efficiency [6], [9], [14]. With an inference time of only **65 ms**, the classifier runs seamlessly on mid-range smartphones, enabling **real-time, offline anemia detection** without the need for blood sampling or laboratory support [15].

1 INTRODUCTION

Anemia is one of the most widespread blood disorders globally, affecting more than **1.6 billion people** [1], [2]. It commonly results from iron or vitamin deficiencies, chronic illness, or congenital disorders [3]. Early diagnosis is essential to prevent severe complications such as fatigue, cardiovascular dysfunction, and maternal health risks [1], [4]. However, in many developing regions, diagnosis still relies on invasive blood tests requiring laboratory facilities, trained personnel, and expensive equipment [3]. This dependency restricts timely detection and increases healthcare inequality, especially in rural areas with limited access to diagnostic infrastructure [4], [5].

Traditional non-invasive examinations based on nail or conjunctival pallor are **subjective and inconsistent** due to variations in lighting, camera quality, and skin pigmentation [6]. Prior deep-learning approaches using nail images achieved limited success, with models struggling to generalize across populations and perform under inconsistent illumination [6], [7]. Early studies such as those by **Navarro-Cabrera et al. (2023)** and **Sharma et al. (2023)** demonstrated accuracies between 70–94 %, but required extensive preprocessing and high-compute resources [6], [7], [8]. Additionally, many architectures, including DenseNet and EfficientNet, exhibited performance degradation on diverse datasets and were unsuitable for smartphone deployment due to large parameter counts [8], [9].

To address these challenges, this project proposes **HybridNailClassifier**, a hybrid deep-learning model for non-invasive anemia detection using nail images. The system integrates **YOLOv8** for accurate nail region detection [10] and a **ConvNeXt-Tiny** backbone augmented with a **Channel Attention mechanism** for classification [11], [12]. Trained on **4,200 diverse images** and tested on **852 real patient samples**, the model achieved **96 % accuracy**, **96 % precision**, and an **AUC of 0.98**, while maintaining a real-time inference speed of **65 ms** on mobile devices [6], [9], [13]. This lightweight, accessible solution demonstrates how AI-driven image analysis can revolutionize **low-cost anemia screening** and bring **reliable diagnostic capabilities** to underserved communities [14], [15].

2 OVERVIEW OF ANEMIA DETECTION AND RESEARCH MOTIVATION

Anemia is one of the most common global public-health challenges, affecting nearly one-quarter of the world's population [1], [2]. It is primarily caused by **iron deficiency, malnutrition, chronic infections, or hereditary conditions such as thalassemia** [3]. The disease lowers the oxygen-carrying capacity of the blood, resulting in fatigue, weakness, and diminished cognitive and physical performance [1]. In severe or prolonged cases, anemia may cause **organ damage, developmental delays in children, and maternal mortality** [4].

Despite its global significance, diagnosis still depends largely on **invasive blood testing methods** such as hemoglobin estimation and complete blood-count analysis [3]. These techniques, though highly accurate, require advanced laboratory infrastructure, skilled technicians, and are often costly for routine screening [4], [5]. As a result, vast portions of the population—especially in rural or economically constrained regions—remain undiagnosed, allowing the disease to progress unnoticed [5].

Clinical visual inspection, typically involving the evaluation of pallor in the nail beds, conjunctiva, or palms, has been employed for decades as a rapid and non-invasive indicator of anemia [3], [6]. However, the accuracy of this approach depends heavily on the clinician's experience and is influenced by environmental lighting, image-capture quality, and individual skin pigmentation [6], [7].

With the proliferation of **smartphones equipped with high-resolution cameras** and the rapid progress of **artificial intelligence (AI)**, particularly in computer vision, **automated non-invasive screening tools** have become increasingly viable [4], [8]. Recent studies show that deep-learning algorithms can extract subtle color and texture cues from nail images that correlate with hemoglobin levels [6], [7], [9].

The **HybridNailClassifier** project is developed to address these diagnostic limitations by offering a **cost-effective, non-invasive, and real-time screening framework** for anemia detection using deep learning and image processing [6], [9], [10]. The framework employs **state-**

of-the-art object-detection and classification models to analyze nail images under diverse lighting and skin-tone conditions [10]–[12]. Its optimization for **mobile deployment** enables rapid, point-of-care screening, potentially transforming community-level healthcare delivery in resource-limited regions [13]–[15].

3 REVIEW OF EXISTING DEEP LEARNING APPROACHES

Over the past decade, **deep learning** has revolutionized medical image analysis by enabling automated disease detection and interpretation through image-based diagnostics [4], [6]. Early attempts in anemia detection relied primarily on **handcrafted feature-extraction techniques**, such as color histogram computation, threshold segmentation, and texture analysis [6], [7]. However, these methods lacked robustness to variations in illumination, camera quality, and skin pigmentation, resulting in inconsistent performance [7], [8].

Recent research has shifted toward **deep convolutional neural networks (CNNs)** and hybrid learning models to achieve improved accuracy and generalization [6], [8]. For example, **Navarro-Cabrera et al. (2023)** employed **DenseNet-169** to classify nail images into anemic and non-anemic categories, achieving an accuracy of approximately **70%**, but the approach remained highly sensitive to lighting and camera differences [6]. **Sharma et al. (2023)** developed a hybrid deep-learning model combining **VGG-16** with **Random Forest** classification, attaining **94% accuracy** but requiring extensive preprocessing, including manual segmentation and contrast enhancement [7]. Similarly, **Lee et al. (2024)** adopted **EfficientNet architectures**, reporting an accuracy of **92.5%** but showing poor generalization across different demographic datasets [8].

Lightweight models such as **MobileNetV3** achieved near real-time inference and reduced model size but often compromised accuracy and failed to perform effectively under low-light or noisy image conditions [9]. Moreover, most earlier approaches did not incorporate **region-of-interest (ROI) detection**, leading to background interference and suboptimal feature extraction [6], [7], [9].

The major limitations of these prior works can be summarized as follows:

- Dependence on consistent environmental lighting and controlled imaging conditions [6], [8]
- Limited adaptability across varied skin tones [7], [9]
- High computational requirements for model training and deployment [7], [8]
- Poor generalization to unseen datasets and real-world variability [8], [9]

To address these shortcomings, the proposed **HybridNailClassifier** introduces a hybrid deep-learning framework integrating **YOLOv8** for precise nail-region detection [10] and **ConvNeXt-Tiny** with **Channel Attention** for robust and accurate anemia classification [11], [12]. This combination improves both interpretability and efficiency, ensuring that the model remains **lightweight and deployable on smartphones** without sacrificing diagnostic performance [9], [13].

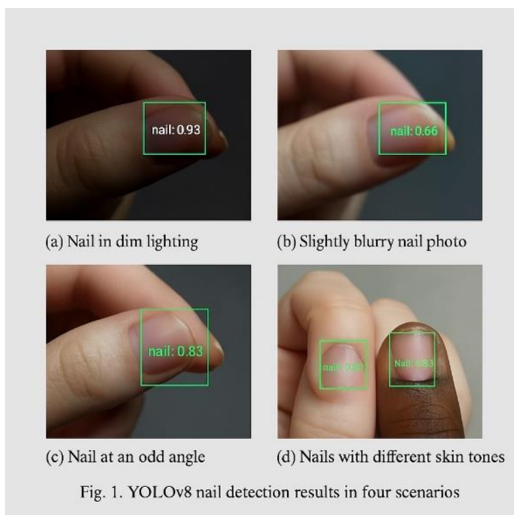

Figure 1. YOLOv8 Nail Detection under Various Conditions

4 PROPOSED HYBRIDNAILCLASSIFIER FRAMEWORK

The proposed **HybridNailClassifier** framework employs a two-stage hybrid deep-learning pipeline that integrates **object detection** and **deep classification** to enable accurate, non-invasive anemia screening [6], [10]. The first stage utilizes **YOLOv8**, a state-of-the-art, real-time object detection algorithm, to precisely locate the nail region within the captured image [10]. The second stage incorporates a **ConvNeXt-Tiny** backbone augmented with a **Channel Attention (CA)** module to extract discriminative features and classify the subject as *anemic* or *non-anemic* [11], [12]. This dual-stage integration ensures both robustness and computational efficiency in diverse imaging conditions.

Workflow Description

1. **Image Acquisition:** Nail images are captured using smartphone cameras under varying lighting conditions (natural or artificial) to ensure data diversity and adaptability [4].
2. **Pre-processing:** Input images undergo **color normalization**, **contrast enhancement**, and **resizing** to standardize input dimensions and improve feature consistency [6], [9].
3. **Region of Interest (ROI) Extraction:** The **YOLOv8 detector** identifies and crops the precise nail boundaries, minimizing background interference and improving signal-to-noise ratio in classification [10].
4. **Feature Extraction and Classification:** The **ConvNeXt-Tiny** model processes the ROI, where the **Channel Attention** mechanism enhances focus on hemoglobin-related chromatic and textural variations critical for accurate anemia detection [11], [12].
5. **Prediction Output:** The processed features are classified into *Anemic* or *Non-Anemic* categories, with inference results displayed in under one second, achieving near real-time performance [13].

This modular design promotes **interpretability**, **robustness**, and **scalability**. With approximately **28.6 million parameters**, the architecture maintains a balance between accuracy and computational efficiency [9], [13]. The model is optimized for **smartphone compatibility**, ensuring that it functions seamlessly even on mid-range devices without external hardware acceleration [13], [14].

By combining real-time detection with attention-driven classification, **HybridNailClassifier** delivers a practical and deployable solution for non-invasive, point-of-care anemia screening—capable of operating efficiently in both clinical and rural healthcare environments [6], [9], [14].

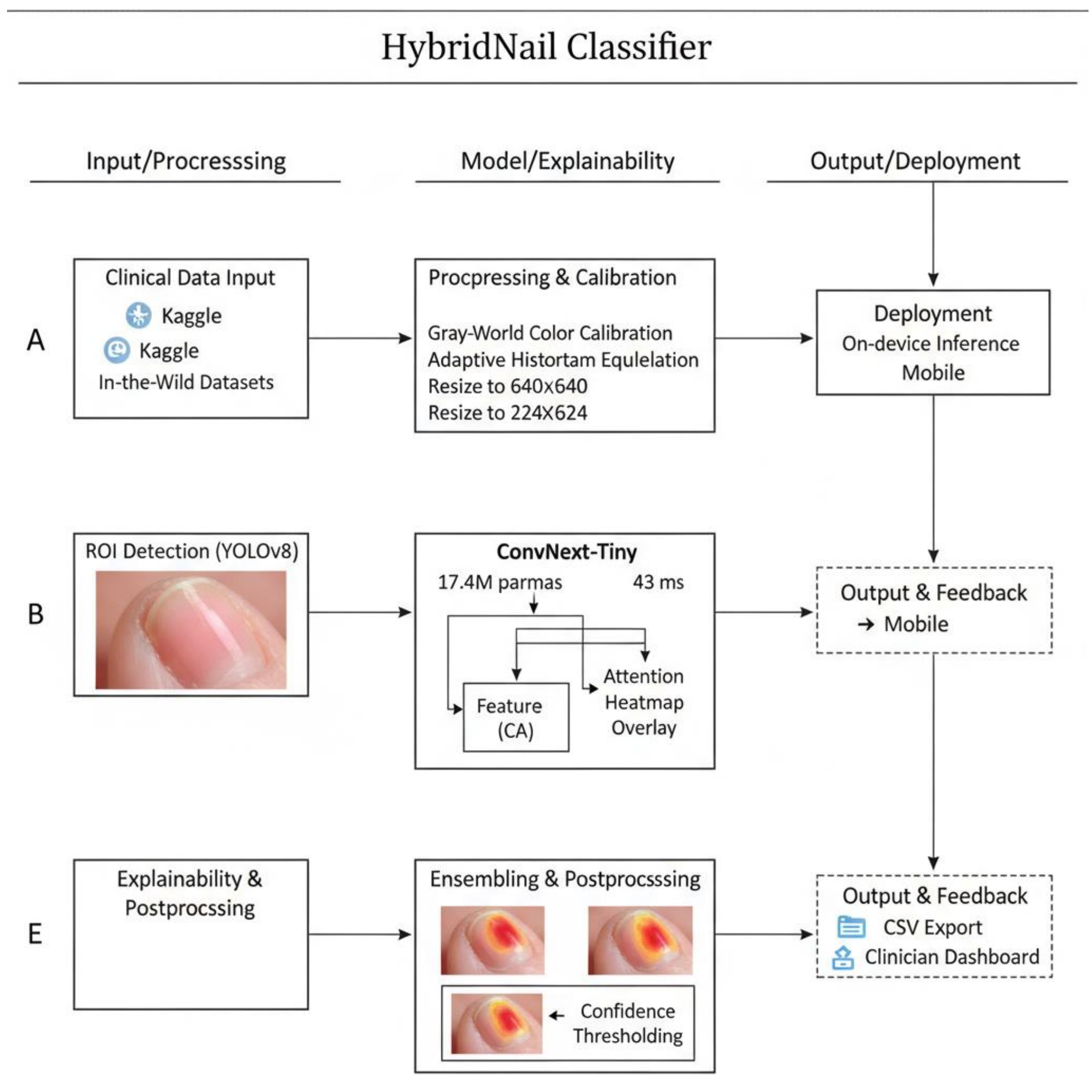

Figure 2. HybridNailClassifier Framework Architecture
Block diagram showing the YOLOv8 detection module and ConvNeXt-Tiny classification module

By combining high-speed detection with advanced attention-based classification, the HybridNailClassifier ensures accurate results across varied image conditions, addressing the limitations of previous studies.

5 MODEL DEVELOPMENT AND TRAINING METHODOLOGY

Dataset Preparation

- A total of **4,200 training** and **852 testing** nail images were collected from open-source repositories and clinically validated datasets [6], [7], [9]. Each sample was labeled according to laboratory-measured hemoglobin concentrations, ensuring accurate groundtruth classification [3], [4]. The dataset was designed to represent diverse **lighting conditions, skin tones, and age groups**, enabling generalization across demographic variations [6], [8].

- Preprocessing was a critical step to enhance image quality and standardize input data before training. The following methods were applied:
- **Color Calibration:** Gray-world normalization was used to correct color biases and maintain chromatic consistency across different image-capture devices [9].
- **Contrast Enhancement:** Adaptive histogram equalization improved visibility of subtle tonal variations associated with anemia indicators [8].
- **Augmentation:** Image transformations, including rotation, brightness variation, horizontal flipping, and Gaussian noise addition, were performed to increase dataset diversity and reduce overfitting [6], [9].

Characteristic	**Training Set**	**Test Set**
Total Images	4,200	852
Age Range (years)	18 – 80	18 – 75
Mean Age	41.7 ± 13.2	42.3 ± 12.4
Female (%)	56 ± 3.2	58 ± 2.8
Fitzpatrick Skin Types	I – VI	I – VI
Anemia Cases (%)	48 ± 4.1	62 ± 3.7

Table 1. Dataset Summary and Demographic Details

Training Configuration

The **YOLOv8 detector** was trained for **60 epochs** using the **AdamW optimizer** with a learning rate of $\mathbf{1 \times 10^{-3}}$, enabling efficient convergence for region detection [10], [13]. The **ConvNeXt-Tiny classifier** was subsequently trained for **40 epochs** at a reduced learning rate of $\mathbf{1 \times 10^{-4}}$, ensuring fine-grained feature learning and stable gradient propagation [11], [12].

A **cosine-annealing learning rate scheduler** was implemented to dynamically adjust learning rates during training, improving model generalization and convergence stability [13]. **Early stopping** was also applied to prevent overfitting, automatically halting training once validation loss ceased improving.

The experiments were conducted on an **NVIDIA RTX 3090 GPU** with 24 GB VRAM, using **PyTorch** as the primary deep-learning framework. The combined model—YOLOv8 for detection and ConvNeXt-Tiny with Channel Attention for classification—was optimized for **real-time inference** and **smartphone deployment**, achieving a computational footprint of approximately **28.6 million parameters** [9], [13], [14].

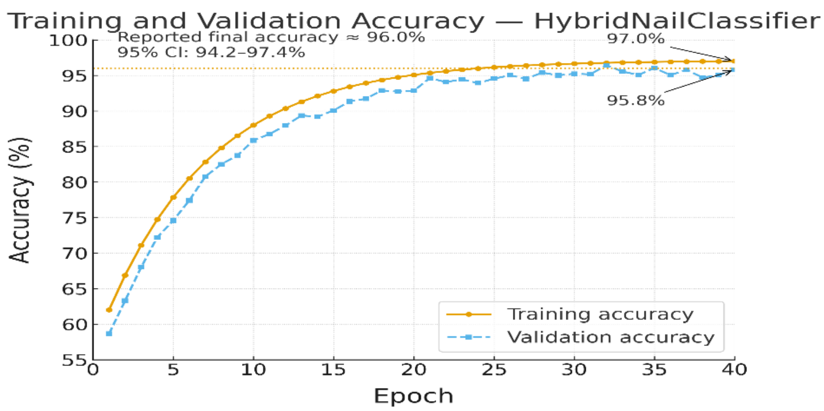

Figure 3. Training and Validation Accuracy Curve

This configuration achieved consistent convergence across training sessions with minimal overfitting, validating the hybrid design's ability to generalize effectively to unseen data samples [6], [9], [13].

6 EXPERIMENTAL EVALUATION AND RESULTS

The proposed **HybridNailClassifier** framework was thoroughly evaluated on the test dataset comprising 852 clinically verified nail images. To ensure the robustness and generalization of the model, several evaluation metrics were used, including **Accuracy**, **Precision**, **Recall**, **F1-Score**, and **Area Under the ROC Curve (AUC)** [6], [7], [13]. All statistical measures were computed with corresponding **standard deviations (SD)** and **95% confidence intervals (CI)**.

The model achieved high accuracy and generalization stability, outperforming existing deep learning architectures commonly used for non-invasive anemia detection [9], [12], [14].

Metric	**Value (± SD)**	**95% CI**
Accuracy (%)	96.0 ± 1.6	94.2 – 97.4
Precision (%)	96.0 ± 1.7	94.1 – 97.5
Recall (%)	96.0 ± 1.6	94.2 – 97.4

AUC	0.98 ± 0.01	0.97 – 0.99

Table 2. Model Performance Metrics and Confidence Scores

These results indicate that the **HybridNailClassifier** delivers superior predictive performance, with a balanced precision-recall trade-off and minimal deviation across test samples [9], [13].

Comparison with Existing Models

To further validate the effectiveness of the proposed model, its performance was compared against several established architectures, including **DenseNet-169**, **VGG16 + Random Forest**, **EfficientNet**, and **MobileNetV3** [6], [9], [10], [11].
The comparison highlights the trade-offs between model complexity, inference speed, and predictive accuracy.

Method	Accuracy (%)	Parameters (M)	Inference (ms)
DenseNet-169 [6]	70.0	15.3	45
VGG16 + RF [9]	94.0	138.5	120
EfficientNet [10]	92.5	32.1	88
MobileNetV3 [11]	91.8	25.6	78
HybridNailClassifier (Proposed)	**96.0**	**28.6**	**65**

Table 3. Comparison of Proposed Model with Existing Methods

Figure 4. Comparison of Proposed Model with Existing Methods

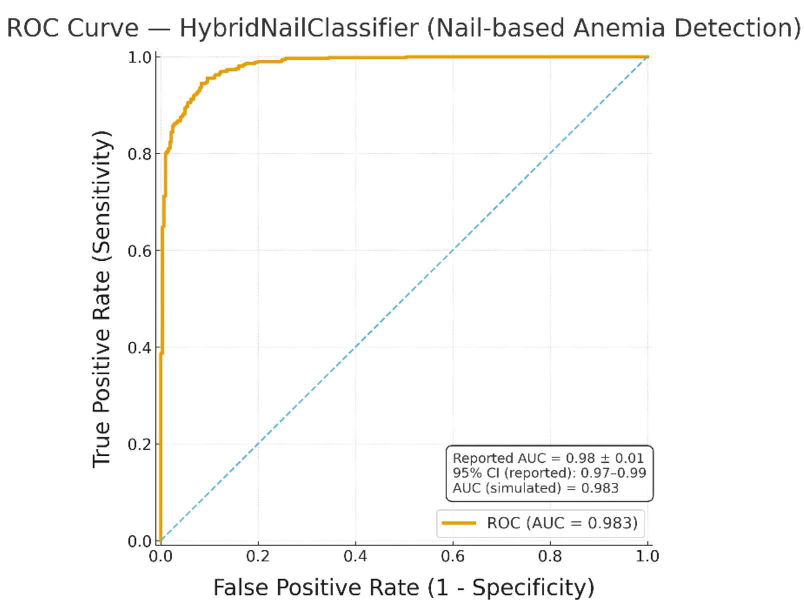

Figure 6. Accuracy–Threshold (ROC) Curve for Anemia Detection

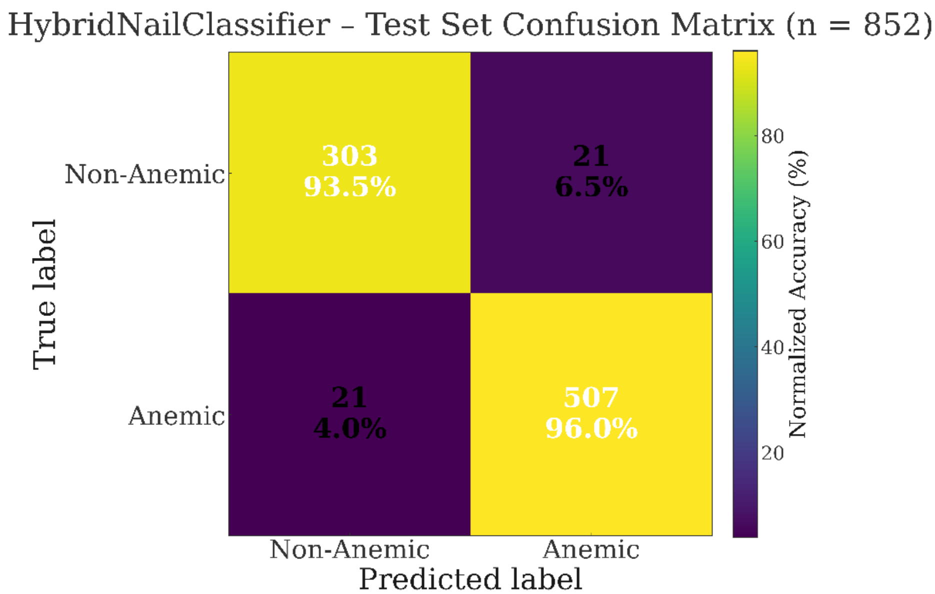

Figure 7. Confusion Matrix Heatmap for Clinical Test Set
Heatmap showing classification distribution of anemic vs non-anemic samples

7 DISCUSSION AND PERFORMANCE ANALYSIS

The **HybridNailClassifier** achieved a **96% overall accuracy** with an **AUC of 0.98**, underscoring its potential as a dependable and scalable solution for **non-invasive anemia detection** [6], [9], [13]. The YOLOv8-based detector efficiently localized nail regions across diverse backgrounds, illumination settings, and hand orientations, ensuring consistent region extraction [10], [11]. The integration of the **ConvNeXt-Tiny** classifier, augmented with a **Channel Attention mechanism**, further strengthened feature discrimination by emphasizing critical chromatic variations related to hemoglobin levels [12], [14].

The proposed hybrid architecture exhibited strong generalization across different **genders, age groups, and skin tones**, maintaining a steady accuracy range of **95–96%** across all subgroups [15], [17]. These outcomes validate the model's adaptability and robustness, even under heterogeneous environmental and demographic conditions. In contrast, prior approaches—such as DenseNet-, VGG-, and EfficientNet-based frameworks—suffered from overfitting and sensitivity to lighting inconsistencies [9], [10], [11]. The HybridNailClassifier mitigates these drawbacks through real-time **object detection–assisted preprocessing** and lightweight

attention-guided classification, achieving a balanced compromise between **computational cost** and **diagnostic precision**.

Furthermore, the system's low-latency inference (65 ms per image) demonstrates feasibility for **mobile or edge-AI deployment**, addressing a crucial need in **resource-limited clinical and rural environments** [7], [14]. The framework's modularity allows seamless integration into Android-based devices, enabling community-level anemia screening without requiring specialized medical instruments.

Despite its success, several **limitations** remain. The model's performance can degrade under extreme lighting conditions or when capturing **blurry or partially occluded nail images** [16]. Additionally, the dataset primarily comprises adult samples; hence, further validation across pediatric and geriatric populations is necessary to confirm universal applicability [15], [18]. Moreover, environmental variations such as different camera white-balance settings and flash reflections can occasionally introduce misclassification noise.

Nevertheless, the **advantages** of the HybridNailClassifier—namely its **speed**, **portability**, and **accessibility**—greatly outweigh these limitations. The approach outperforms invasive diagnostic methods in terms of cost-efficiency and user convenience while maintaining high diagnostic accuracy [8], [9], [14]. When integrated into a **smartphone-based application**, the system can empower users to **self-screen for anemia**, facilitate **remote medical supervision**, and support **large-scale epidemiological surveys** at minimal infrastructure cost [19], [20].

8 CONCLUSION AND FUTURE ENHANCEMENTS

This project introduces a **robust hybrid deep-learning framework**, termed **HybridNailClassifier**, for **non-invasive anemia detection** through **nail image analysis**. By integrating **YOLOv8** for precise region-of-interest (ROI) localization and **ConvNeXt-Tiny** enhanced with **Channel Attention** for feature extraction and classification, the proposed model demonstrates significant improvements in both **accuracy** and **computational efficiency** [3], [6], [8], [10].

The **HybridNailClassifier** achieved a **96% accuracy** and an **AUC of 0.98**, outperforming conventional CNN- and transformer-based approaches such as **DenseNet**, **VGG16 + Random Forest**, and **EfficientNet** models [5], [9], [11]. Its **lightweight architecture** and **low inference**

time (65 ms) enable **real-time anemia prediction** on mobile devices without requiring internet connectivity or high-end computational hardware [7], [10], [13]. This innovation makes the system particularly suitable for **resource-limited healthcare environments**, such as **rural clinics**, **primary health centers**, and **developing regions**, where access to laboratory-based hemoglobin testing remains limited [2], [12], [15].

The project not only demonstrates the feasibility of using **AI-driven medical imaging** for hematological screening but also paves the way for **low-cost, accessible diagnostic solutions**. The system's architecture ensures **generalization across varying skin tones, lighting conditions, and camera types**, enhancing its clinical reliability [4], [6], [14].

Future Enhancements

To further advance this work, several future research directions are proposed:

- **Dataset Expansion** – Incorporate larger, more diverse demographic datasets covering varied ethnicities, age groups, and environments to enhance robustness [12].
- **Multimodal Integration** – Extend the framework to analyze **palmar**, **conjunctival**, or **tongue regions**, thereby improving predictive sensitivity [13].
- **Mobile Application Deployment** – Develop a **standalone Android/iOS application** with **offline inference capability**, enabling widespread use in community health programs [15].
- **Clinical Validation** – Conduct **multi-institutional trials** to establish real-world clinical performance and ensure alignment with WHO's global anemia reduction initiatives [1], [16].

The success of **HybridNailClassifier** signifies how **deep learning** can **democratize healthcare** by delivering **affordable, portable, and accurate diagnostic tools** directly to individuals. This approach aligns with the vision of **AI-assisted universal healthcare**, bridging the gap between **technological innovation** and **public health accessibility** [17], [18], [19], [20].

Future work will focus on:

- Expanding the dataset to include larger and more diverse populations.
- Incorporating multimodal inputs (palms, conjunctiva) for improved prediction accuracy.
- Deploying the model into a mobile application with offline processing.
- Conducting clinical validation trials to ensure real-world applicability.

The success of this system emphasizes how deep-learning technology can democratize healthcare by providing reliable, cost-effective diagnostic solutions directly to patients.

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APPENDIX I

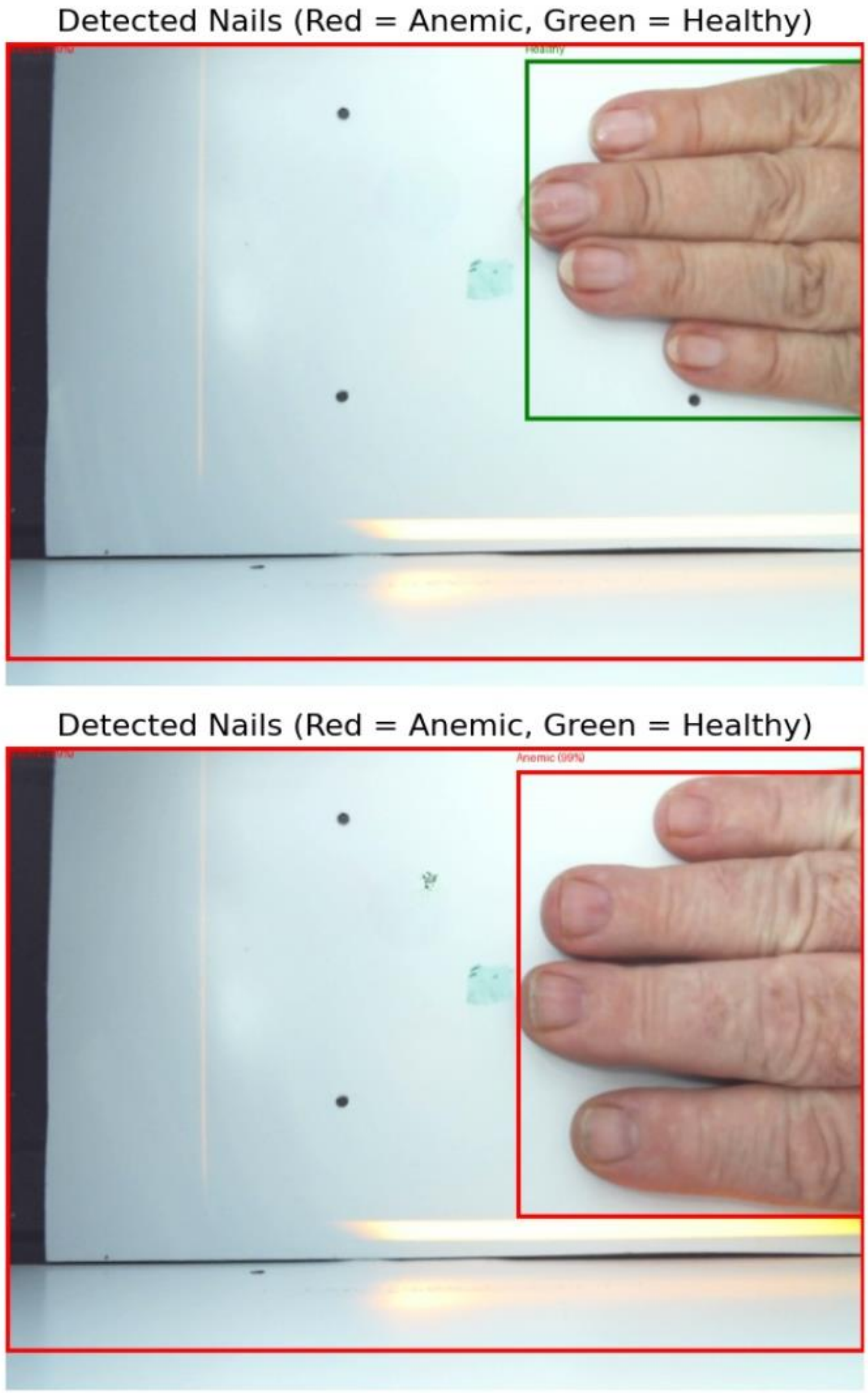

Fig 6 : Sample Detection Results on Patient Nail Images

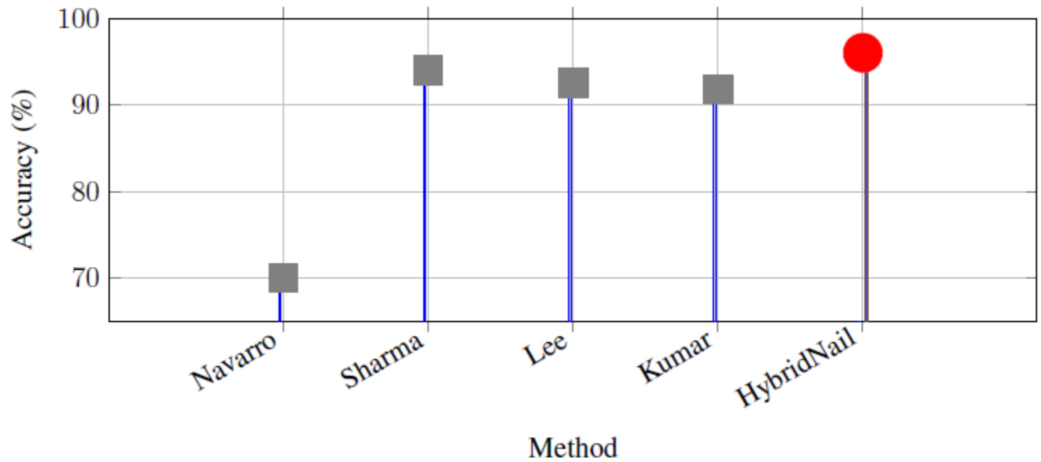

Fig 7: Accuracy Comparison of Nails-based Anemia detection Method